SPOKESBOT ANALYTICS

Product Requiremnets Document (PRD)

PROJECT	SpokesBot Intelligent Analytics Console
TYPE	Web Application
PREPARED BY	Aditya Chaubey
MVP DURATION	**10 Weeks**
TARGET LAUNCH	June 10, 2026
TECH STACK	React+Next.js / Python (FastAPI) / PostgreSQL

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Document Control

Version	Date	Author	Description of Changes
v0.1		Aditya Chaubey	Initial Draft (Analytics Console Scope)
v1.0		Aditya Chaubey	Finalized MVP Scope (Internal Tool, SQL Tracking, Chatbot)
v1.1		Aditya Chaubey	Future Scopes

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1 Project Timeline & Milestones

The MVP development is scheduled for an accelerated 11-week timeline, broken down into 4 key phases.

Phase	Duration	Key Deliverables
Phase 1: Discovery	Week 1	• Finalize PRD, Dashboard UI/UX wireframes. • Set up fresh development environment. • Supabase DB Schema & data ingestion logic.
Phase 2: Core Dev	Weeks 2-5	• **Frontend (Client):** Global date filters, dynamic KPI cards, charts, and Chatbot UI. • **Frontend (Admin):** Internal CSV upload portal (No validation; strict CSV formatting required). • **Backend:** API development for data ingestion, aggregation, and user auth.
Phase 3: Integration	Weeks 6-8	• **AI Insights & Chatbot:** Integrate LLM for text-based summaries and natural language dashboard queries.
Phase 4: QA & UAT	Weeks 9-10	• **Data Testing:** Data accuracy testing against raw CSVs. • **Load Testing:** Simulated concurrent load with large datasets. • **Beta Launch:** Limited release to internal stakeholders.

2 Stakeholders

Role	Description
User (Client)	Logs into the console to interact with metrics, toggle date filters, query the chatbot, and read AI-generated insights.
Admin (Platform Owner)	Pre-formats and uploads data (CSVs) for individual clients, manages client accounts, monitors platform health, and tracks system usage.

3 Key Features (MVP Scope)

3.1 A. The Interactive Dashboard (Client-Facing)

✓ **Included in MVP:**

- **Global Time/Date Filter Dropdown:** A sticky control at the top of the console allowing users to slice data by standard presets (*Today, Yesterday, This Week, Last 7 Days, This Month, Last 30 Days, YTD*, and *Custom Date Range*).
- **Dynamic KPI Metric Cards:** A top row of clickable, high-level metric tiles (e.g., *Total Revenue, Active Users, Conversion Rate*). Clicking a card sets it as the active metric and automatically redraws the main charts.
- **Instant UI Reactivity:** When a user clicks a new date range or toggles a metric card, charts and the AI Insights widget refresh instantly without a page reload.
- **Conversational Chatbot ("SpokesBot"):** Integrated natural language data assistant allowing users to query their dashboard directly (e.g., "Why did sales drop in May?") to dynamically generate charts and contextual answers.
- **Automated AI Insights:** A dedicated widget that uses Gemini to analyze the currently filtered data and generate plain-text summaries.

× **Explicitly Excluded from MVP:**

- No Data Export (PDF/CSV) functionality.
- No Real-time streaming database ingestion (Data is batched/uploaded).
- No Predictive modeling or forecasting algorithms.

Scenario Logic (MVP)

Scenario	Behavior
User changes dropdown from "This Month" to "Today"	Frontend instantly filters the dataset. Charts re-render, KPI numbers update, and the AI widget generates a new daily summary.
User asks chatbot: "What was the highest performing day last week?"	Chatbot parses intent, queries the dataset, and responds with the specific date and metric visualization.

3.2 B. Admin Portal & Data Ingestion (Internal)

✓ **Included in MVP:**

- **Data Upload Interface:** A secure, internal-only page where the Admin can upload CSV files.
- **Client Mapping:** Dropdown to select which `org_id` (Client) the uploaded dataset belongs to.
- **Direct Data Ingestion:** System assumes CSVs are perfectly formatted by the Admin prior to upload (No frontend validation safety net).

Scenario Logic (Admin)

Scenario	Behavior
Admin uploads perfectly formatted "Q1_Sales.csv" for "Client A"	System parses the CSV directly into the database under Client A's `org_id`, updating Client A's dashboard instantly.

4 Tech Stack & Architecture

Layer	Tech Choice
Frontend	React, Next.js 14 (App Router), TypeScript, Tailwind CSS, Recharts
State Management	Zustand or React Context (for instant global filter reactivity)
Backend/API	**FastAPI (Python)**
Data Processing	Python (Pandas/NumPy)
Backend / Auth	Supabase (Auth, DB, Storage)
Database	PostgreSQL
AI Engine	**OpenAI / Gemini 3.1 Pro (for generating text insights and chatbot responses from structured data)**

5 Event Tracking (SQL-Based)

✓ **Dashboard & Chatbot Events Tracked Directly via SQL:**

- `console_login` (Session Start)
- `date_filter_changed` (Tracks which timeframes users care about most)
- `kpi_card_clicked` (Tracks which metrics are most interacted with)
- `chatbot_query` (Tracks user questions to SpokesBot for continuous improvement)
- `insight_generated` (Tracking AI summary usage)

6 Data Dictionary & Schema Specifications

6.1 Core Tables (Supabase/PostgreSQL)

Table: `organizations` (Internal Client Isolation)

Column	Type	Description
`id`	UUID	Primary Key.
`name`	TEXT	Client / Company name.

Table: `datasets` (Uploaded Data)

Column	Type	Description
`id`	UUID	Primary Key.
`org_id`	UUID	Foreign Key to `organizations`.
`source_name`	TEXT	e.g., "Q1_Sales_Data.csv".
`raw_data`	JSONB	The structured data payload.
`uploaded_at`	TIMESTAMPTZ	Timestamp of ingestion.

Table: `ai_insights` (Cached Summaries & Chat Logs)

Column	Type	Description
`id`	UUID	Primary Key.
`dataset_id`	UUID	Foreign Key to `datasets`.
`filter_context`	TEXT	e.g., "Last 7 Days - Revenue".
`insight_text`	TEXT	The LLM-generated summary or chatbot reply.
`generated_at`	TIMESTAMPTZ	When the insight/reply was created.

7 Fallbacks & Failover Logic

Scenario	What to Implement	Why It Matters
LLM API Failure	Hide "AI Insights" widget and gracefully disable the Chatbot input with a "Temporarily unavailable" banner. Charts still function.	Ensures core analytics value remains intact even if AI APIs are down.
Large Dataset Rendering	If client-side filtering slows browser, implement data downsampling before passing to Recharts.	Prevents frontend freezing and janky UI transitions when switching dates.

8 Testing & QA Strategy

Test Category	Methodology	Expected System Behavior
1. UI Reactivity	Toggle rapidly between "Today", "This Week", and "This Month" etc.	Charts must redraw instantly without flickering or page reloads.
2. Data Accuracy	Compare DB queries against raw CSV uploads using a specific date filter.	Dashboard metrics must perfectly match the source data for that exact timeframe.
3. Tenant Isolation	Attempt to fetch `org_id=2` data using an `org_id=1` token.	Supabase Row Level Security strictly blocks the query.
4. AI Grounding	Chatbot query: "Why did sales drop?"	Chatbot answer must accurately reflect the data trends and not hallucinate reasons not present in the dataset.
5. Admin Routing	Admin uploads a CSV and assigns it to Client A.	Verify the data *only* appears on Client A's dashboard and remains completely invisible to Client B.

9 Future Roadmap (Post-MVP)

Feature	Description
Data Export & Reporting	PDF and CSV export capabilities for clients.
Admin Validation Net	Automated error catching and UI validation for Admin CSV uploads.

10 Stakeholder Sign-off

By signing below, the stakeholders agree that this document represents the complete scope for the MVP phase of the **Intelligent Analytics Console**. Any additions after this date will be treated as Change Requests.

Product Owner Date:	**Lead Architect** Date:
Executive Sponsor (CEO) Date:	**Engineering Lead** Date: